

Data Science with Python – Module 4 Exercise

Hands-on Exercise No. 4

Course: Data Science with Python

Module: AI & Model Deployment

Total Marks: 10

Due Date: 27/06/2026

Instructions

Prepare your solution in MS Word.

Submit in **.doc** or **.docx** format.

Late or copied submissions will not be accepted.

Learning Outcome

After completing this exercise, you will be able to:

- Understand deep learning and AI model concepts.
 - Build and evaluate a simple neural network model.
 - Save and load trained machine learning models.
 - Deploy a model for basic prediction tasks.
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Problem Statement

You are working as a Junior Data Scientist for an AI startup. The client wants a simple AI-based prediction system that can classify whether a customer is likely to purchase a product based on historical data. Your task is to build, save, and simulate deployment of a basic AI model using Python.

Task Requirements

Create a new Jupyter Notebook or Python script.

Perform the following steps:

1. Load Dataset

- Import a dataset (e.g., customers.csv) using Pandas.

- Display first 10 rows and check dataset structure.
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2. Data Preprocessing

- Handle missing values.
 - Encode categorical variables.
 - Normalize numerical features if required.
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3. Build AI Model

- Use a simple Neural Network model using **TensorFlow / Keras**.
 - Define input layer, hidden layers, and output layer.
 - Compile the model using appropriate optimizer and loss function.
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4. Train Model

- Split dataset into training and testing sets (80/20).
 - Train the model and display accuracy/loss graphs.
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5. Model Evaluation

- Evaluate model performance on test data.
 - Display accuracy score and confusion matrix.
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6. Save & Load Model

- Save trained model using `model.save()`.
 - Load the saved model and perform a test prediction.
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7. Simple Deployment Simulation

- Create a function that takes user input and returns prediction (Purchase: Yes/No).
 - Demonstrate prediction with at least 2 sample inputs.
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Visualizations Required

- Training accuracy vs loss graph
 - Confusion matrix
 - Prediction output examples
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Submission Requirements

Take screenshots of:

- Code implementation
 - Model training process
 - Evaluation results
 - Prediction output
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Submission Guidelines

Upload your **.docx file** with code, screenshots, explanations, and results via LMS.